Topic 7: Envelope Theorems and Value Functions

(How the value of an optimal choice changes when the problem changes)

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1. Why this topic belongs here

Topic 5 taught us to separate two objects that come out of an optimization problem:

- the **argmax**, which tells us which choices are optimal;
- the **value function**, which tells us how good the best attainable outcome is.

Topic 6 then gave us derivatives, gradients, first-order conditions, and total differentiation. We used those tools to ask how an optimal *choice* changes with a parameter. The Implicit Function Theorem, which we will use in Topic 8, follows exactly that route: differentiate the first-order condition and solve for the derivative of the optimizer.

But economists often want a different object:

> "When a price, a resource, or a type changes, how does the **optimized value** change?"

That question appears almost immediately in demand theory. The indirect utility function, expenditure function, and profit function are all value functions. It appears again in mechanism design, where an agent's equilibrium utility is the value of choosing the best report, and in dynamic programming, where the value function records the payoff from an optimal continuation plan.

The **envelope theorem** answers this question. Its basic message is:

> "To differentiate the optimized value, hold the optimal choice fixed and differentiate the objective directly with respect to the parameter."

That sounds surprising. The optimal choice usually changes with the parameter, so why can we ignore that change? We start with the smooth interior case, where the answer comes from the first-order condition. We then add constraints, where Lagrange multipliers record the value of moving a boundary. Finally, we turn to Milgrom and Segal's theorem, which shows that the envelope formula can survive even when the choice set is arbitrary and the optimizer jumps.

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2. Parameterized optimization and the value function

Let X be a choice set, let θ be a parameter, and consider

$$V(\theta) = \max_{x \in X} f(x, \theta).$$

The function f is the **primitive objective**: it tells us the payoff from a particular choice x at a particular parameter θ.

The function V is the **value function**: it tells us the best payoff attainable at each θ.

The set of maximizers is

$$X^*(\theta) = \text{argmax}_{x \in X} f(x, \theta).$$

If the maximizer is unique, we may write it as a function $x^*(\theta)$. Then

$$V(\theta) = f(x^*(\theta), \theta).$$

These are different objects. The derivative $x^{*\prime}(\theta)$ describes **choice comparative statics**. The derivative $V'(\theta)$ describes **value comparative statics**. The Implicit Function Theorem is typically used for the first. The envelope theorem is built for the second.

For example, if x is a firm's input and θ is the output price, then:

- $x^*(\theta)$ tells us how input demand changes with the price;
- $V(\theta)$ is maximized profit;
- $V'(\theta)$ tells us the marginal value of a price increase.

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3. The differentiable interior envelope theorem

We begin with the cleanest version.

Theorem (Differentiable interior envelope theorem).
Let

$$V(\theta) = \max_{x \in X} f(x, \theta),$$

where $X \subseteq \mathbb{R}^n$ does not depend on θ. Fix an open interval of parameter values. Suppose on that interval:

1. f is continuously differentiable in (x, θ);
2. the optimizer $x^*(\theta)$ is unique, interior, and differentiable in θ;

3. the first-order condition characterizes the optimum:

$$\nabla_x f(x^*(\theta), \theta) = 0.$$

Then V is differentiable and

$$V'(\theta) = \frac{\partial f}{\partial \theta}(x^*(\theta), \theta).$$

> "The derivative of the optimized value equals the direct effect of the parameter on the objective, evaluated at an optimal choice. We do not need the derivative of the optimal choice."

Proof

Start from the identity

$$V(\theta) = f(x^*(\theta), \theta).$$

The chain rule gives

$$V'(\theta) = \nabla_x f(x^*(\theta), \theta) \cdot x^{*\prime}(\theta) + \frac{\partial f}{\partial \theta}(x^*(\theta), \theta).$$

The first term records the indirect effect of θ through the changing optimizer. But the first-order condition says

$$\nabla_x f(x^*(\theta), \theta) = 0.$$

Therefore

$$V'(\theta) = 0 \cdot x^{*\prime}(\theta) + \frac{\partial f}{\partial \theta}(x^*(\theta), \theta),$$

which is the claimed formula. □

Vector parameters

If the parameter is a vector $\theta = (\theta_1, \ldots, \theta_k)$, apply the same argument one component at a time:

$$\nabla_\theta V(\theta) = \nabla_\theta f(x^*(\theta), \theta).$$

Each entry of the gradient answers one ceteris-paribus question: how does optimized value change when that component of the environment changes slightly?

4. Why the indirect effect disappears

The proof is short, but the intuition deserves more space.

Suppose θ rises a little. Two things happen:

1. **Direct effect:** even if the agent kept choosing the old $x^*(\theta)$, the payoff $f(x^*(\theta), \theta)$ would change because θ entered the objective.
2. **Reoptimization effect:** the agent changes the choice from $x^*(\theta)$ to $x^*(\theta + \mathrm{d}\theta)$.

The envelope theorem says that the reoptimization effect is negligible **to first order**. Why? At an interior optimum, the objective is locally flat in every choice direction:

$$\nabla_x f(x^*(\theta), \theta) = 0.$$

Moving the choice a small amount therefore has no first-order payoff effect. Reoptimization still matters for the level of the value function, and it may matter at second order, but it drops out of the derivative.

This is sometimes summarized by saying that "the choice effect is second order." That phrase is useful only if we remember what makes it true: the relevant first-order condition must hold. At a boundary, or when the feasible set itself moves with the parameter, the missing first-order term can return through the constraint.

A small example

Consider

$$V(\theta) = \max_{x \in \mathbb{R}} \left\{ \theta x - \frac{x^2}{2} \right\}.$$

The first-order condition is

$$\theta - x = 0,$$

so $x^*(\theta) = \theta$. Substituting gives

$$V(\theta) = \frac{\theta^2}{2}, \qquad V'(\theta) = \theta.$$

The envelope formula obtains the same answer without differentiating x^*:

$$\frac{\partial}{\partial \theta}\left(\theta x - \frac{x^2}{2}\right)\Big|_{x=x^*(\theta)} = x^*(\theta) = \theta.$$

For this quadratic problem, solving for x^* was easy. In larger problems, the envelope theorem is valuable precisely because the optimizer may be difficult or impossible to write in closed form.

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5. Constrained optimization and KKT

Most economic optimization problems are constrained, and the constraint may itself depend on the parameter. A consumer's budget moves with prices and wealth. A firm's technology may move with productivity. A mechanism must satisfy incentive and participation constraints.

Consider the maximization problem

$$\begin{aligned} V(\theta) = \max_x \quad & f(x,\theta) \\ \text{subject to} \quad & g_j(x,\theta) \geq 0, \qquad j = 1,\dots,m, \\ & h_k(x,\theta) = 0, \qquad k = 1,\dots,r. \end{aligned}$$

An inequality constraint is **binding** (or active) at x^* if $g_j(x^*,\theta) = 0$. It is slack if $g_j(x^*,\theta) > 0$.

The Lagrangian

The **Lagrangian** collects the objective and the constraints in one function:

$$\mathcal{L}(x,\lambda,\mu;\theta) = f(x,\theta) + \sum_{j=1}^{m} \lambda_j g_j(x,\theta) + \sum_{k=1}^{r} \mu_k h_k(x,\theta),$$

where $\lambda_j \geq 0$ for the inequality constraints and the equality multipliers μ_k can have either sign.

Why is this useful? At an unconstrained interior optimum, $\nabla_x f = 0$. At a constrained optimum this usually fails: the objective may still point uphill, but the constraint prevents us from walking in that direction. The Lagrangian asks whether the objective's gradient can be balanced by a weighted combination of the constraint gradients.

The weights are the **Lagrange multipliers**. They record how strongly each binding constraint pushes back against the objective. Equality constraints can push in either direction, so μ_k can have either sign. An inequality of the form

$g_j \geq 0$ can only prevent movement toward $g_j < 0$, which is why its multiplier has the sign restriction $\lambda_j \geq 0$ under our maximization convention.[1]

The Karush–Kuhn–Tucker conditions

Under a suitable **constraint qualification**, an optimum satisfies the **Karush–Kuhn–Tucker (KKT) conditions**:

1. **Stationarity**

$$\nabla_x f + \sum_j \lambda_j \nabla_x g_j + \sum_k \mu_k \nabla_x h_k = 0.$$

2. **Primal feasibility**

$$g_j(x^*, \theta) \geq 0, \qquad h_k(x^*, \theta) = 0.$$

3. **Dual feasibility**

$$\lambda_j \geq 0.$$

4. **Complementary slackness**

$$\lambda_j g_j(x^*, \theta) = 0.$$

Complementary slackness says that a slack constraint has multiplier zero. A constraint can carry a positive shadow value only when it binds.

Read the four conditions together:

- **Stationarity:** after attaching the correct shadow value to every binding constraint, there is no remaining first-order direction of improvement.
- **Primal feasibility:** the candidate choice actually satisfies the original constraints.
- **Dual feasibility:** inequality constraints carry multipliers with the correct sign.
- **Complementary slackness:** either a constraint binds or its multiplier is zero (possibly both); a slack constraint has no marginal value.

The first condition alone is not KKT. A point can make the Lagrangian stationary while violating the original problem. The other three conditions keep the Lagrangian connected to feasible choice.

[1] Sign conventions differ across books. If a maximization constraint is instead written as $q_j(x, \theta) \leq 0$, the corresponding term is $-\lambda_j q_j$ with $\lambda_j \geq 0$. Nothing economic changes. Pick one convention and use it consistently.

Why a constraint qualification appears

KKT multipliers do not automatically exist for every way of writing every constraint. Consider the problem

$$\max_x x \quad \text{subject to} \quad -x^2 \geq 0.$$

The only feasible point is $x^* = 0$, so it is certainly optimal. But the gradient of the constraint is $-2x^* = 0$, while the gradient of the objective is 1. No multiplier can make

$$1 + \lambda(-2x^*) = 0.$$

The feasible set is perfectly clear, but its local description is degenerate: the constraint has a zero gradient exactly where we need it to describe the boundary.

One standard way to rule this out is the **linear independence constraint qualification (LICQ)**: the gradients (with respect to x) of the equality constraints and the binding inequality constraints must be linearly independent at the optimum. Under LICQ, a local optimum has KKT multipliers, and those multipliers are unique.

There are other constraint qualifications. For a convex problem, a common one is **Slater's condition**: there exists a feasible point at which every inequality constraint is strictly satisfied. We do not need a catalogue. The economic message is that a constraint qualification makes sure the equations describe the local boundary faithfully enough for multipliers to exist.

Necessary versus sufficient

KKT is first a **necessary-condition theorem**:

> "At a regular local optimum, there must exist multipliers satisfying the KKT conditions."

This identifies candidates. It does not, by itself, prove that a candidate is optimal—just as $\nabla f = 0$ did not prove optimality in Topic 6.

KKT becomes sufficient under the appropriate global shape restrictions. With our maximization convention, suppose:

- $f(\cdot, \theta)$ is concave;
- every $g_j(\cdot, \theta)$ is concave, so each set $\{x : g_j(x, \theta) \geq 0\}$ is convex;
- every equality constraint $h_k(\cdot, \theta) = 0$ is affine.

Then the feasible set is convex, and any feasible point with multipliers satisfying KKT is a global maximizer. If f is strictly concave, the maximizer is unique.

Thus:

- regularity makes KKT **necessary**;
- concavity and convex feasibility make KKT **sufficient**.

A corner-solution example

Consider

$$\max_{x \geq 0} \left\{ \theta x - \frac{x^2}{2} \right\}.$$

Write the constraint as $g(x) = x \geq 0$. The Lagrangian is

$$\mathcal{L}(x, \lambda; \theta) = \theta x - \frac{x^2}{2} + \lambda x.$$

KKT requires

$$\theta - x + \lambda = 0, \qquad x \geq 0, \qquad \lambda \geq 0, \qquad \lambda x = 0.$$

There are two cases.

1. If $\theta > 0$, the optimum is interior: $x^* = \theta > 0$. Complementary slackness forces $\lambda^* = 0$, and stationarity becomes the familiar unconstrained first-order condition.
2. If $\theta < 0$, the optimum is the corner $x^* = 0$. Stationarity then gives $\lambda^* = -\theta > 0$. The multiplier records how strongly the nonnegativity constraint prevents the agent from following the unconstrained optimum into negative x.

At $\theta = 0$, $x^* = 0$ and $\lambda^* = 0$: the constraint binds but has zero marginal value. This is why complementary slackness says $\lambda g = 0$, not "binding if and only if the multiplier is positive."

> "A multiplier is the marginal value of relaxing a constraint. KKT tells us how that value enters the first-order conditions."

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6. The constrained envelope theorem

We can now state the version used throughout demand and production theory.

Theorem (Differentiable constrained envelope theorem).
Consider the parameterized maximization problem in Section 5. Suppose the objective and constraints are continuously differentiable, an optimal solution and associated KKT multipliers exist, and the solution and relevant multipliers

vary differentiably near the parameter value under consideration. Suppose also that the locally relevant active constraints remain active; standard regularity conditions deliver this for constraints with nonzero multipliers. Then

$$V'(\theta) = \frac{\partial \mathcal{L}}{\partial \theta}\big(x^*(\theta), \lambda^*(\theta), \mu^*(\theta); \theta\big).$$

Equivalently,

$$V'(\theta) = f_\theta + \sum_{j=1}^{m} \lambda_j^* g_{j,\theta} + \sum_{k=1}^{r} \mu_k^* h_{k,\theta},$$

with every term evaluated at the optimum and the current parameter.

> "Differentiate the Lagrangian with respect to the parameter while holding the choices and multipliers fixed at their optimal values."

Why the formula is true

Differentiate the value:

$$V'(\theta) = \nabla_x f(x^*, \theta) \cdot x^{*\prime}(\theta) + f_\theta(x^*, \theta).$$

Stationarity lets us replace the gradient of the objective:

$$\nabla_x f = -\sum_j \lambda_j^* \nabla_x g_j - \sum_k \mu_k^* \nabla_x h_k.$$

For a binding inequality with a nonzero multiplier, local continuity of the multiplier and complementary slackness keep the constraint binding nearby. Differentiating

$$g_j(x^*(\theta), \theta) = 0$$

gives

$$\nabla_x g_j \cdot x^{*\prime}(\theta) + g_{j,\theta} = 0.$$

For a slack inequality, or a binding inequality with a zero multiplier at the parameter under study, $\lambda_j^* = 0$, so it contributes nothing to the formula. Equality constraints satisfy the same differentiated identity with h_k in place of g_j.

Substituting these identities into $V'(\theta)$ leaves

$$V'(\theta) = f_\theta + \sum_j \lambda_j^* g_{j,\theta} + \sum_k \mu_k^* h_{k,\theta} = \mathcal{L}_\theta.$$

The optimizer still moves, but KKT converts the value of that movement into the shadow values of the moving constraints.

Minimization problems

The same rule holds for minimization: form the Lagrangian with the sign convention appropriate to the constraints, then differentiate that Lagrangian directly with respect to the parameter. For example, for

$$\min_x c(x, \theta) \quad \text{subject to} \quad q(x, \theta) \leq 0,$$

use

$$\mathcal{L}(x, \lambda; \theta) = c(x, \theta) + \lambda q(x, \theta), \qquad \lambda \geq 0,$$

and again $V'(\theta) = \mathcal{L}_\theta$.

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7. Why the constraint term cannot be ignored

Consider

$$V(\theta) = \max_x x \quad \text{subject to} \quad x \leq \theta.$$

The optimum is $x^*(\theta) = \theta$, so $V(\theta) = \theta$ and $V'(\theta) = 1$.

But the objective $f(x, \theta) = x$ does not directly depend on θ, so $f_\theta = 0$. The unconstrained envelope formula would give the wrong answer.

Write the constraint as $g(x, \theta) = \theta - x \geq 0$. The Lagrangian is

$$\mathcal{L}(x, \lambda; \theta) = x + \lambda(\theta - x).$$

Stationarity gives $1 - \lambda = 0$, so $\lambda^* = 1$. The constrained formula gives

$$V'(\theta) = \mathcal{L}_\theta = \lambda^* = 1.$$

This is exactly the economic interpretation of the multiplier:

> "The multiplier is the marginal value of one more unit on the right-hand side of the constraint."

The example also shows why Topic 6's interior first-order condition is not enough for constrained problems. At the solution, $f_x = 1$, not 0. The boundary prevents us from walking in the profitable direction, and the multiplier records how valuable it would be to move that boundary.

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8. Three workhorses from demand and production theory

The constrained envelope theorem is the common engine behind results that often appear under different names.

8.1 Shephard's lemma

The expenditure function is

$$e(p, \bar{u}) = \min_x p \cdot x \quad \text{subject to} \quad u(x) \geq \bar{u}.$$

Let $h(p, \bar{u})$ denote Hicksian demand. Use the Lagrangian

$$\mathcal{L}(x, \mu; p, \bar{u}) = p \cdot x + \mu\big(\bar{u} - u(x)\big), \qquad \mu \geq 0.$$

Suppose the expenditure minimizer is regular and unique and the expenditure function is differentiable at $(p, \bar{u})$.[2]

Prices enter only through $p \cdot x$. Therefore

$$\frac{\partial e(p, \bar{u})}{\partial p_\ell} = \frac{\partial \mathcal{L}}{\partial p_\ell} = x_\ell^* = h_\ell(p, \bar{u}).$$

In vector form:

$$\nabla_p e(p, \bar{u}) = h(p, \bar{u}).$$

> "The price gradient of the minimum bill is the compensated demand bundle."

8.2 Roy's identity

The indirect utility function is

$$v(p, w) = \max_x u(x) \quad \text{subject to} \quad w - p \cdot x \geq 0.$$

[2] Uniqueness is a convenient sufficient condition for the gradient formula, not the most general one. With several Hicksian bundles and a nondifferentiable expenditure function, the corresponding statement is set-valued and uses supergradients.

The Lagrangian is

$$\mathcal{L}(x, \lambda; p, w) = u(x) + \lambda(w - p \cdot x).$$

Suppose the optimum is regular, the budget binds, and v is differentiable at (p, w). Local nonsatiation is a standard condition ensuring that the budget binds. If demand is unique, write it as $x(p, w)$.

The constrained envelope theorem gives

$$\frac{\partial v}{\partial w} = \lambda^*, \qquad \frac{\partial v}{\partial p_\ell} = -\lambda^* x_\ell^*.$$

Dividing the second identity by the first gives Roy's identity:

$$x_\ell(p, w) = -\frac{\partial v(p, w)/\partial p_\ell}{\partial v(p, w)/\partial w},$$

provided $\partial v/\partial w = \lambda^* > 0$.

The denominator is not a mysterious correction. It converts utility units into wealth units.

8.3 Hotelling's lemma

The profit function is

$$\pi(p) = \max_{y \in Y} p \cdot y,$$

where the production set Y does not depend on p. Prices enter the objective directly, so

$$\frac{\partial \pi(p)}{\partial p_\ell} = y_\ell^*(p).$$

In vector form:

$$\nabla_p \pi(p) = y(p)$$

whenever the relevant derivative exists and the optimal netput is unique.

Shephard, Roy, and Hotelling are not three unrelated tricks. They are three readings of the same envelope formula.

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9. Where the smooth proof becomes fragile

The chain-rule proof in Section 3 assumed much more than the formula itself seems to need:

- the choice set was a well-behaved subset of Euclidean space;
- the optimizer was interior;
- the optimizer was unique and differentiable;
- the primitive objective was differentiable in the choice.

Mechanism design routinely violates these assumptions. A message set can be discrete or otherwise arbitrary. Several reports may be optimal. The optimal report may jump when the agent's type crosses a cutoff. At exactly such a cutoff, the value function may have a kink.

Consider

$$V(t) = \max_{x \in \{-1,1\}} xt = |t|.$$

For $t < 0$, the optimizer is $x^*(t) = -1$. For $t > 0$, it is $x^*(t) = 1$. At $t = 0$, both choices are optimal. The optimizer jumps, and V is not differentiable at 0.

This is not a failure of economics. It is correct information:

- the right derivative is $V'_+(0) = 1$;
- the left derivative is $V'_-(0) = -1$;
- there is no single derivative at the kink.

Any general envelope theorem must respect this. It should not claim differentiability where the value function genuinely has none. What it can do is characterize the derivative wherever it exists, characterize one-sided derivatives under additional conditions, and provide an integral representation when the value function is sufficiently regular.

That is the contribution of Milgrom and Segal.

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10. The Milgrom–Segal envelope formula

Let X now be an **arbitrary set**. It need not be convex, finite-dimensional, or even a topological space. Let the scalar parameter be $t \in [0, 1]$, and define

$$V(t) = \sup_{x \in X} f(x, t),$$

with optimal-choice correspondence

$$X^*(t) = \{x \in X : f(x, t) = V(t)\}.$$

Whenever we invoke an optimizer, we assume $X^*(t)$ is nonempty.

Theorem (Milgrom–Segal envelope formula).
Fix $t \in (0,1)$ and an optimizer $x^* \in X^*(t)$. Suppose the partial derivative $f_t(x^*, t)$ exists.

- If V has a right derivative at t, then

$$V'_+(t) \geq f_t(x^*, t).$$

- If V has a left derivative at t, then

$$V'_-(t) \leq f_t(x^*, t).$$

- If V is differentiable at t, then

$$V'(t) = f_t(x^*, t).$$

The last equality holds for **every** optimizer $x^* \in X^*(t)$ whose parameter derivative exists.

The striking part is what is absent:

- no derivative with respect to x;
- no first-order condition;
- no convexity of X;
- no continuity or differentiability of an optimizer $x^*(t)$.

Proof: freeze the old optimizer

Because x^* is optimal at t,

$$f(x^*, t) = V(t).$$

At a nearby parameter t', the value $V(t')$ is the supremum over all choices, so it is at least the payoff from keeping the old optimizer:

$$f(x^*, t') \leq V(t').$$

Subtracting gives

$$f(x^*, t') - f(x^*, t) \leq V(t') - V(t).$$

If $t' > t$, divide by $t' - t > 0$ and take $t' \downarrow t$:

$$f_t(x^*, t) \leq V'_+(t).$$

If $t' < t$, division reverses the relevant comparison and taking $t' \uparrow t$ gives

$$V'_-(t) \leq f_t(x^*, t).$$

When V is differentiable, $V'_-(t) = V'_+(t) = V'(t)$, so the two inequalities force

$$V'(t) = f_t(x^*, t).$$

□

This proof gives a different intuition from the smooth proof. We do not say that the changing optimizer has a second-order effect. We do not differentiate the optimizer at all. We simply compare the value function with the payoff from keeping yesterday's optimal choice available today.

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11. Kinks, multiple optimizers, and directional derivatives

Theorem 10 tells us that if V is differentiable, all optimizers must generate the same direct parameter effect. Therefore, a useful diagnostic is:

> "If two optimal choices have different values of f_t, expect a kink in the value function."

Under familiar compactness and continuity assumptions this statement becomes exact. Suppose X is compact, $f(\cdot, t)$ is upper semicontinuous in x, and $f_t(x, t)$ is continuous in (x, t). Then:

$$V'_+(t) = \max_{x \in X^*(t)} f_t(x, t), \qquad V'_-(t) = \min_{x \in X^*(t)} f_t(x, t).$$

Consequently, V is differentiable at t if and only if $f_t(x, t)$ has the same value for every $x \in X^*(t)$.

Return to $V(t) = \max_{x \in \{-1,1\}} xt$. At $t = 0$, both -1 and 1 are optimal, and

$$f_t(-1, 0) = -1, \qquad f_t(1, 0) = 1.$$

The minimum is the left derivative and the maximum is the right derivative. The kink is exactly the disagreement among optimal branches.

This also clarifies the role of uniqueness. A unique optimizer is a convenient sufficient condition for avoiding such disagreement, but uniqueness alone is not the deep requirement. The value function can be differentiable with several optimizers, provided all optimal branches have the same parameter derivative.

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12. A short bridge: absolute continuity and "almost everywhere"

Mechanism design usually needs more than a derivative at one point. It needs to recover utility differences between types by integrating the derivative:

$$V(t) - V(s) = \int_s^t V'(r)\,\mathrm{d}r.$$

Differentiability almost everywhere is not, by itself, enough for this identity. Even continuity is not enough.[3]

We need two pieces of language.

Almost everywhere

A subset $N \subseteq [0,1]$ has **measure zero** if, for every $\varepsilon > 0$, it can be covered by countably many intervals whose total length is less than ε. A statement holds **almost everywhere** if it can fail only on a set of measure zero.

Finite and countable sets have measure zero (countable sets are defined in Topic 1, Section 5). Thus, saying that a formula holds almost everywhere allows isolated kinks, and even countably many kinks, without treating them as economically important regions.

Absolute continuity

A function $F : [0,1] \to \mathbb{R}$ is **absolutely continuous** if, for every $\varepsilon > 0$, there exists $\delta > 0$ such that for every finite collection of disjoint intervals (a_i, b_i),

$$\sum_i (b_i - a_i) < \delta \quad \Longrightarrow \quad \sum_i |F(b_i) - F(a_i)| < \varepsilon.$$

Absolute continuity rules out a function accumulating a nontrivial change on a set of arbitrarily small total length.

The fundamental fact we need is:

[3] The Cantor function is continuous, has derivative zero almost everywhere, and yet rises from 0 to 1. Therefore it cannot be recovered by integrating its derivative. The missing property is absolute continuity.

> "An absolutely continuous function is differentiable almost everywhere, its derivative is integrable, and the function equals its starting value plus the integral of its derivative."

Thus, if F is absolutely continuous,

$$F(t) = F(0) + \int_0^t F'(s)\,\mathrm{d}s.$$

Topic 15 develops measure and the Lebesgue integral formally. For the present theorem, this one-dimensional bridge is enough to say precisely what the integral representation requires.

Integration by parts

The fundamental fact has a consequence that we need whenever an integral contains a product. It is the product rule of differentiation, read backwards.

Recall the product rule from school: at every point s where F and G are both differentiable, so is their product, and

$$(FG)'(s) = F'(s)G(s) + F(s)G'(s).$$

Now let F and G be absolutely continuous on $[0,1]$. Each of them is differentiable almost everywhere, so the product rule holds almost everywhere.[4] The product FG is again absolutely continuous.[5] So the fundamental fact applies to FG. Both integrals below exist, because F' and G' are integrable and F and G are bounded:

$$F(1)G(1) - F(0)G(0) = \int_0^1 F'(s)G(s)\,\mathrm{d}s + \int_0^1 F(s)G'(s)\,\mathrm{d}s.$$

Solving for one of the two integrals gives **integration by parts**:

$$\int_0^1 F(s)G'(s)\,\mathrm{d}s = \Big[F(s)G(s)\Big]_0^1 - \int_0^1 F'(s)G(s)\,\mathrm{d}s,$$

where $\big[F(s)G(s)\big]_0^1$ is shorthand for $F(1)G(1) - F(0)G(0)$.

[4] The points where F is not differentiable form a set of measure zero, and so do the points where G is not. Two sets of measure zero together still have measure zero: cover each of them by intervals of total length less than $\varepsilon/2$.

[5] Both functions are continuous (apply the definition with a single interval), hence bounded on $[0,1]$ by the Weierstrass theorem of Topic 4, Section 8, say $|F| \le M$ and $|G| \le M$. Since $F(b)G(b) - F(a)G(a) = F(b)\big(G(b) - G(a)\big) + G(a)\big(F(b) - F(a)\big)$, the change of FG over any interval is at most M times the sum of the changes of F and of G. A small total length therefore forces a small total change of the product.

In English:

> "Inside an integral we may move the derivative from one factor to the other. The price is a minus sign and a boundary term."

Nothing here is special to $[0,1]$: on any interval $[a,b]$ the same formula holds with a and b in place of 0 and 1. The rule earns its keep when one factor is easier to differentiate than to integrate, typically because that factor is itself defined as an integral. Section 14 has exactly such a case.[6]

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13. Milgrom–Segal's integral envelope theorem

We now add conditions that make the value function absolutely continuous.

Theorem (Milgrom–Segal integral envelope theorem).
Let X be nonempty and

$$V(t) = \sup_{x \in X} f(x,t), \qquad t \in [0,1].$$

Suppose:

1. $V(0) < \infty$;
2. $f(x,\cdot)$ is absolutely continuous for every $x \in X$;
3. there exists an integrable function $b : [0,1] \to \mathbb{R}_+$ such that, for almost every t,

$$|f_t(x,t)| \le b(t)$$

for every $x \in X$. Thus, the exceptional measure-zero set does not depend on the choice x.

Then V is finite-valued and absolutely continuous.

If, in addition, $f(x,\cdot)$ is differentiable for every x and $X^*(t)$ is nonempty almost everywhere, choose $x^*(t) \in X^*(t)$ wherever the optimizer set is nonempty and choose any element of X on the remaining measure-zero set. Then

$$V(t) = V(0) + \int_0^t f_t(x^*(s), s)\, \mathrm{d}s.$$

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[6] For continuously differentiable F and G this is in every calculus text, for example Rudin, *Principles of Mathematical Analysis* (third edition), Theorem 6.22. Integration is not part of MWG's Mathematical Appendix; the product rule is stated there in Section M.A.

"Even if optimal choices jump, the value function can be recovered by integrating the direct parameter effect along any optimal selection."

Where the bound is used

Because $V(0) < \infty$, the uniform bound first guarantees that $V(t)$ is finite for every t. Indeed, for every x,

$$f(x,t) \le f(x,0) + \int_0^t b(s)\,\mathrm{d}s,$$

so taking suprema gives

$$V(t) \le V(0) + \int_0^t b(s)\,\mathrm{d}s < \infty.$$

For $t' < t''$,

$$|V(t'') - V(t')| \le \sup_{x\in X} |f(x,t'') - f(x,t')|.$$

Because each branch $f(x,\cdot)$ is absolutely continuous,

$$|f(x,t'') - f(x,t')| \le \int_{t'}^{t''} |f_t(x,s)|\,\mathrm{d}s \le \int_{t'}^{t''} b(s)\,\mathrm{d}s.$$

The same bound works for every x, so

$$|V(t'') - V(t')| \le \int_{t'}^{t''} b(s)\,\mathrm{d}s.$$

This uniform control makes V absolutely continuous. Therefore V is differentiable almost everywhere and can be reconstructed from its derivative. At every differentiability point, the formula from Section 10 identifies that derivative as $f_t(x^*(t),t)$.

Why the integrable bound is not cosmetic

Let $X = (0,\infty)$ and

$$f(x,t) = g(t/x), \qquad g(z) = ze^{-z}, \qquad t \in [0,1].$$

For every fixed x, the function $f(x,\cdot)$ is smooth. For $t > 0$, choosing $x = t$ makes $t/x = 1$, the unique maximizer of g, so

$$V(t) = g(1) = e^{-1}.$$

At $t = 0$, however, $f(x, 0) = g(0) = 0$ for every x, so $V(0) = 0$. The value function jumps at zero.

What failed? The parameter derivatives cannot be bounded uniformly by an integrable function near zero. Indeed, after writing $z = t/x$,

$$|f_t(x, t)| = \frac{ze^{-z}|1 - z|}{t},$$

whose supremum over x is a positive constant times $1/t$. That bound is not integrable near zero. Smoothness of each individual branch is not enough; we need control that is uniform across the choices over which we take the supremum.

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14. Mechanism design: the payoff identity

The Milgrom–Segal theorem was built for settings like mechanism design.

Suppose a one-dimensional type $t \in [\underline{t}, \bar{t}]$ can choose any message m from an arbitrary message set M. A message produces allocation probability $q(m)$ and payment $p(m)$. With quasilinear utility,

$$f(m, t) = tq(m) - p(m).$$

The agent's equilibrium utility is the value function

$$U(t) = \sup_{m \in M} \{tq(m) - p(m)\}.$$

The choice set M may be discrete. The optimal message may jump. None of this affects the direct type derivative:

$$f_t(m, t) = q(m).$$

If the mechanism implements an allocation rule $q^*(t) = q(m^*(t))$ and the boundedness conditions of Section 13 hold, then

$$U'(t) = q^*(t) \quad \text{for almost every } t,$$

and

$$U(t) = U(\underline{t}) + \int_{\underline{t}}^{t} q^*(s)\,\mathrm{d}s.$$

This is the **payoff identity** (or envelope characterization) used in one-dimensional mechanism design.

It has two immediate consequences.

Incentive compatibility implies monotonicity

If truth-telling is optimal, then for any $t > s$,

$$U(t) \geq tq^*(s) - p^*(s),$$

and

$$U(s) \geq sq^*(t) - p^*(t).$$

Using $U(r) = rq^*(r) - p^*(r)$ and adding the inequalities gives

$$(t-s)\big(q^*(t) - q^*(s)\big) \geq 0.$$

Therefore $q^*(t) \geq q^*(s)$: higher types receive weakly higher allocations.

Allocation plus one utility level pins down payments

Since

$$p^*(t) = tq^*(t) - U(t),$$

the payoff identity gives

$$p^*(t) = tq^*(t) - U(\underline{t}) - \int_{\underline{t}}^{t} q^*(s)\,\mathrm{d}s.$$

Once the allocation rule and the lowest type's utility are fixed, the payment rule is fixed. This is the mathematical core of **revenue equivalence**: mechanisms that implement the same allocation rule and give the same baseline utility generate the same expected payments.

In a first-price auction, the same logic appears with

$$f(b,t) = (t-b)G(b),$$

where $G(b)$ is the probability of winning with bid b. The direct type derivative is $G(b)$. Therefore the derivative of the bidder's optimized payoff is her equilibrium winning probability, even though the optimal bid itself changes with type.

Adding up over types: integration by parts at work

A seller cares about the payments she collects in total, across all the types she might face. Computing that total is a standard use of the integration by parts of Section 12.

Let $\underline{t} = 0$ and $\bar{t} = 1$, and suppose there is a population of buyers whose types are spread evenly over $[0, 1]$.[7] Total payments are $\int_0^1 p^*(t)\,\mathrm{d}t$. Write $R(t) := \int_0^t q^*(s)\,\mathrm{d}s$ for the integral in the payment formula above. Integrating that formula term by term,

$$\int_0^1 p^*(t)\,\mathrm{d}t = \int_0^1 tq^*(t)\,\mathrm{d}t - U(0) - \int_0^1 R(t)\,\mathrm{d}t.$$

The last term is an integral of an integral, and integration by parts removes one layer. By the payoff identity, $R(t) = U(t) - U(0)$. Since U is absolutely continuous (Section 13) with $U' = q^*$ almost everywhere, R is absolutely continuous as well, with $R' = q^*$ almost everywhere and $R(0) = 0$. For the second factor we need an absolutely continuous function H with $H'(t) = 1$, and we are free to choose the constant: $H(t) = -(1 - t)$ is a straight line, hence absolutely continuous (take $\delta = \varepsilon$), with $H'(t) = 1$ and $H(1) = 0$. Integration by parts as in Section 12, with $F = R$ and $G = H$ there, gives

$$\begin{aligned}\int_0^1 R(t)\,\mathrm{d}t = \int_0^1 R(t)H'(t)\,\mathrm{d}t &= \Big[R(t)H(t)\Big]_0^1 - \int_0^1 R'(t)H(t)\,\mathrm{d}t \\ &= \int_0^1 (1-t)\,q^*(t)\,\mathrm{d}t.\end{aligned}$$

The boundary term vanishes at both ends, because $H(1) = 0$ and $R(0) = 0$. That is why we chose $-(1 - t)$ rather than t.

The result has a reading. By the payoff identity, the allocation given to type t shows up in the utility of every type above t, and a share $1 - t$ of the population lies above t.

Putting the pieces together,

[7] In the language of Topic 9: the type has the cumulative distribution function $\Phi(t) = t$ on $[0, 1]$ (the uniform distribution), and the total below is the expected payment. For a general distribution with cumulative distribution function Φ and density $\varphi > 0$ (we write Φ and φ because F and f are taken in this topic), the same steps go through with $\varphi(t)\,\mathrm{d}t$ in place of $\mathrm{d}t$ and $H(t) = -(1 - \Phi(t))$ in place of $-(1 - t)$, and the term $2t - 1$ below becomes $t - \frac{1-\Phi(t)}{\varphi(t)}$.

$$\int_0^1 p^*(t)\,\mathrm{d}t = \int_0^1 (2t-1)\,q^*(t)\,\mathrm{d}t - U(0).$$

> "Allocating to type t brings in t, but it obliges the seller to leave $1-t$ to the types above. What she keeps is $2t-1$."

For $t < \frac{1}{2}$ this is negative: serving a low type costs the seller more in utility left to higher types than it brings in. The term $2t-1$ is the simplest instance of what auction theory calls a **virtual value**.

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15. Value functions beyond static demand

The same structure reappears in dynamic programming. There, the "choice" can be an entire continuation plan, and the parameter can be the initial state. The direct Milgrom–Segal formulation requires a common plan set across initial states. This is available when feasibility is parameter-independent or when the problem can be reparameterized over a common set of plans. If changing the initial state changes which plans are feasible, the parameter enters the constraints and the constrained-envelope formula—including its multiplier terms—is the appropriate version.

With that qualification, the smooth envelope theorem says: if the optimal plan and value function are differentiable, differentiate the primitive payoff with respect to the initial state while holding the plan fixed. Milgrom–Segal says something more robust: the envelope formula may remain valid at differentiability points even if the identity of the optimal plan changes discontinuously.

This is why envelope arguments appear in:

- Bellman equations and comparative statics of value functions;
- optimal stopping, where the optimal stopping date can jump;
- contract theory, where the chosen contract can change at a cutoff;
- auctions, where allocation rules can be discontinuous.

The common object is always the same:

$$\text{value at parameter} = \sup_{\text{feasible choices}} \text{payoff from that choice at the parameter.}$$

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16. Where this appears in MWG and beyond

The classical material is developed in:

- **MWG, Mathematical Appendix M.J** (unconstrained maximization),
- **MWG, Mathematical Appendix M.K** (constrained maximization),

- **MWG, Mathematical Appendix M.L** (the envelope theorem),
- **MWG, Chapter 3** (indirect utility, expenditure, Shephard's lemma, and Roy's identity),
- **MWG, Chapter 5** (profit and cost functions, including Hotelling's lemma).

The arbitrary-choice-set theorem is not in MWG. The primary reference is:

> Milgrom, P. and I. Segal (2002), "Envelope Theorems for Arbitrary Choice Sets," *Econometrica* 70(2), 583–601, https://doi.org/10.1111/1468-0262.00296.

Their Theorem 1 gives the envelope formula at differentiability points; Theorem 2 gives the absolute-continuity and integral representation used in mechanism design. The paper also develops directional derivatives, parameterized constraints, saddle-point problems, and optimal stopping applications.

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17. What you should take away

After this topic, you should be comfortable with:

- distinguishing a policy or argmax from a value function;
- deriving the smooth interior envelope formula by the chain rule and first-order condition;
- reading KKT multipliers as shadow values and using

$$V_\theta = \mathcal{L}_\theta$$

  in constrained problems;
- recognizing Shephard's lemma, Roy's identity, and Hotelling's lemma as applications of one theorem;
- understanding why a kink can arise when optimal branches have different parameter derivatives;
- stating the Milgrom–Segal formula without assuming a smooth optimizer or a convex choice set;
- using absolute continuity to turn an almost-everywhere derivative into an integral payoff identity;
- using integration by parts to move a derivative from one factor of an integrand to the other;
- seeing why the mechanism-design identity

$$U(t) = U(\underline{t}) + \int_{\underline{t}}^{t} q(s)\,\mathrm{d}s$$

is an envelope theorem.

The most useful habit is simple: whenever you see a parameterized maximization problem, identify both outputs. Ask separately how the optimizer changes and how the optimized value changes. The first question may require the Implicit Function Theorem or monotone comparative statics. The second is the territory of envelope theorems.